Original Article

Inflammatory and Atherogenic Markers as Predictors of Coronary Artery Disease in Patients with Suspected Coronary Artery Disease Undergoing Coronary Angiography

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Abstract

Background: **Coronary artery disease (CAD) remains a leading cause of death worldwide, with inflammation playing a key role in its pathophysiology. Inflammatory biomarkers may offer valuable insights into disease mechanisms and risk stratification, potentially informing therapeutic decisions.**

Objectives: **This study aimed to investigate the association between key inflammatory markers — C-reactive protein (CRP), neutrophil-to-lymphocyte ratio (NLR), and platelet-to-lymphocyte ratio (PLR) — and both the presence and severity of coronary artery stenosis (CAS) in patients with suspected CAD undergoing coronary angiography.**

Methods: **This retrospective study included 767 patients referred for coronary angiography due to suspected CAD. The primary outcome was the presence of significant CAS. Demographic data, medical history, and blood samples were collected. Statistical analyses included descriptive methods and regression modeling as well as receiver operating characteristic (ROC) curve analysis to determine optimal cutoff values for biomarkers, with area under the curve (AUC) used to assess predictive performance. A p-value < 0.05 was considered statistically significant.**

Results: **Of the 767 patients (mean age 58.3±12.4 years; 56.7% men), 496 had significant CAS (≥ 50%) and 271 had nonsignificant CAS (< 50%). Levels of CRP (0.44 vs. 0.13 mg/dL, p = 0.01) and NLR (2.6 vs. 2.0, p = 0.05) were significantly higher in patients with CAS. No significant difference was observed for PLR (129.4 vs. 118.6, p = 0.13). In multivariable analysis, CRP (OR = 1.188, p = 0.044) and NLR (OR=1.567, p = 0.03) were identified as independent predictors of significant CAS. However, ROC curve analysis showed limited discriminative ability for both CRP (AUC = 0.526) and NLR (AUC = 0.510), with values only slightly above chance.**

Conclusion: **Increased levels of CRP and NLR are significantly associated with CAS and may serve as potential biomarkers for detecting CAD.**

Keywords: **Coronary Artery Disease; C-Reactive Protein; Biomarcadores.**

Introduction

Coronary artery disease (CAD) remains a leading cause of morbidity and mortality worldwide, primarily due to the progression of atherosclerosis and its associated inflammatory mechanisms.[1] Identifying reliable biomarkers for the early detection and progression of coronary artery stenosis (CAS) — a key clinical manifestation of CAD — is essential to enhance diagnostic precision and inform therapeutic strategies.[2]

Systemic inflammation and lipid dysregulation are well-established contributors to CAD pathogenesis.[3] Biomarkers such as C-reactive protein (CRP) have been widely investigated for their ability to reflect inflammatory activity linked to atherosclerosis.[4] In addition, novel composite indices — e.g., the neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), systemic immune-inflammation index (SII), and atherogenic index of plasma (AIP) — have emerged as potential indicators of cardiovascular risk.[5-6] These markers, derived from routine laboratory tests, offer a cost-effective, accessible means of evaluating inflammation and lipid-related cardiovascular risk in clinical practice.[7]

Despite growing interest, the predictive value of these inflammatory and atherogenic markers for identifying the presence and severity of CAS remains under active investigation.[8] Moreover, the influence of demographic and clinical variables — including sex-based differences — on these biomarkers is not yet fully understood.[9] Negative findings in this area are particularly valuable, as they help prevent premature clinical adoption of ineffective markers

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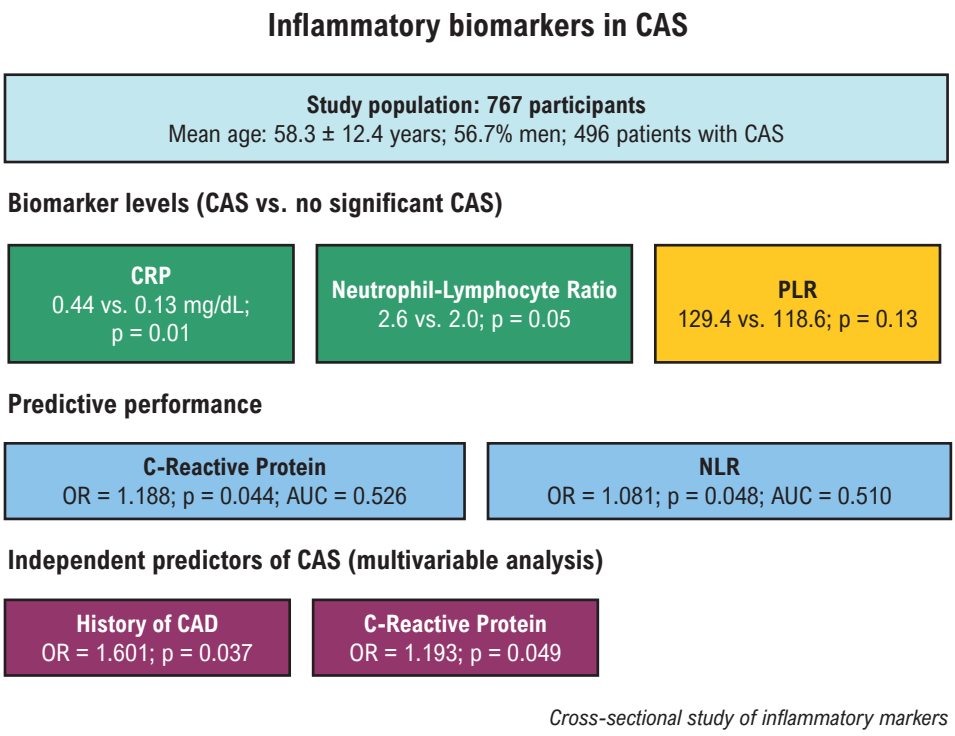
Central Illustration: Inflammatory and Atherogenic Markers as Predictors of Coronary Artery Disease in Patients with Suspected Coronary Artery Disease Undergoing Coronary Angiography



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Inflammatory and Atherogenic Markers as Predictors of Coronary Artery Disease in Patients with Suspected Coronary Artery Disease Undergoing Coronary Angiography. AUC: area under the curve; CAD: coronary artery disease; CAS: coronary artery stenosis; CRP: C-reactive protein; NLR: neutrophil-to-lymphocite ratio; PLR: platelet-to-lymphocite ratio; OR: odds ratio.

and redirect research efforts toward more promising diagnostic tools.

This retrospective observational study seeks to address these knowledge gaps by evaluating the relationship between inflammatory and atherogenic markers and CAS in patients with suspected CAD undergoing coronary angiography. By analyzing a range of biomarkers — including CRP, NLR, PLR, SII, and AIP — and comparing their predictive value across sexes, this study offers a comprehensive assessment of their utility in detecting significant CAS. The findings aim to enhance the clinical understanding and application of these biomarkers in CAD risk stratification and decision-making.[10]

Methods

Study design

This retrospective observational study was conducted to evaluate the association between inflammatory and atherogenic biomarkers and the presence of CAS. The study was carried out at a tertiary care center between January 2022 and December 2023, including patients who underwent coronary angiography for suspected CAD during this period.

Study population

All consecutive patients who underwent coronary angiography for suspected CAD between January 2022 and December 2023 were screened for eligibility. Inclusion required the availability of complete clinical, laboratory, and angiographic data. Exclusion criteria were active infection; systemic inflammatory or autoimmune disease; malignancy; acute coronary syndrome (to avoid confounding elevations in inflammatory markers due to myocardial injury); recent use (within 1 month) of anti-inflammatory or immunosuppressive drugs; history of organ transplantation; or incomplete medical records.

A total of 767 patients met the eligibility criteria and were included in the final analysis. Demographic data, comorbidities, and lifestyle factors — including smoking status — were systematically collected. All data were extracted from the hospital's electronic health records and verified by trained research personnel to ensure accuracy and reliability.

Data collection and laboratory assessments

Data collection encompassed three domains: i) demographic and clinical characteristics; ii) laboratory biomarkers; and iii) angiographic findings. Demographic variables included age,

sex, and smoking status. Clinical data comprised comorbidities such as hypertension, diabetes mellitus, and a history of CAD.

Blood samples were obtained prior to coronary angiography. Complete blood counts were used to determine neutrophil, lymphocyte, and platelet counts. Biochemical analyses included serum levels of CRP, total cholesterol, high-density lipoprotein cholesterol (HDL-C), low-density lipoprotein cholesterol (LDL-C), and triglycerides.

Derived inflammatory and lipid-related markers were calculated as follows: i) NLR: absolute neutrophil count divided by absolute lymphocyte count, reflecting systemic inflammation; ii) PLR: platelet count divided by lymphocyte count, indicating the balance between thrombosis and inflammation; iii) systemic immune-inflammation index (SII): (neutrophil count × platelet count) / lymphocyte count, representing a comprehensive immune-inflammatory profile; iv) AIP: log(triglycerides / HDL-C), serving as an indicator of lipid-related cardiovascular risk.

All laboratory analyses were performed by using standardized protocols on automated analyzers in the hospital's central laboratory.

Coronary angiography

Coronary angiography was conducted using standard fluoroscopic techniques via femoral or radial arterial access. Procedures were performed by experienced interventional cardiologists who were blinded to the patients' clinical and laboratory information. The degree of CAS was visually assessed, with significant CAS defined as ≥ 50% luminal narrowing in any major epicardial coronary artery. Patients were then categorized into two groups: those with significant CAS (≥ 50%) and those without (< 50%), enabling comparative and regression analyses.

Subgroup analysis

A subgroup analysis was conducted in order to explore potential sex-related differences in inflammatory and atherogenic biomarkers. Men and women were compared with respect to CRP, NLR, PLR, SII, and AIP values. This analysis aimed to determine whether sex influenced the expression of these biomarkers.

Ethical considerations

The study protocol was reviewed and approved by the institutional ethics committee, in accordance with the ethical standards outlined in the 2024 revision of the World Medical Association Declaration of Helsinki. Due to the retrospective design of the study, the requirement for informed consent was waived. Patient confidentiality was safeguarded by anonymizing all data prior to analysis, and access to sensitive information was restricted to authorized study personnel.

Quality control and reliability

All measurements were conducted using calibrated equipment under standardized operating conditions in order to ensure the reliability and consistency of data. The hospital's electronic health records system was audited to minimize data extraction errors. Laboratory tests were repeated when initial results fell outside expected ranges or showed discrepancies. Readings were confirmed by 2 independent cardiologists through consensus in order to reduce interobserver variability in angiographic interpretation.

Statistical analysis

Descriptive statistics were used to summarize patient characteristics. Continuous variables were expressed as mean ± standard deviation (SD) for normally distributed data or as median and interquartile range for non-normally distributed data. Categorical variables were reported as frequencies (percentages).

Group comparisons (CAS vs. no CAS) were conducted using chi-square test or Fisher's exact test for categorical variables abd independent samples *t*-test (for normally distributed continuous variables) or Mann-Whitney *U* test (for non-normally distributed variables).

Comparisons between men and women were also performed using the independent samples *t*-test or the Mann-Whitney *U* test, depending on data distribution. Normality of continuous variables was assessed using the Shapiro-Wilk test and visual inspection of histograms.

Univariable and multivariable logistic regression analyses were conducted to identify associations between predictor variables and CAS. Results were reported as odds ratios (ORs) with 95% CIs. Variables with a p-value < 0.10 in univariable analysis, along with clinically relevant covariates (i.e., age, sex, CRP, NLR, LDL-C, hypertension, diabetes, and smoking status), were included in the multivariable model.

Receiver operating characteristic (ROC) curve analysis was used to determine optimal cutoff values for CRP and NLR, with area under the curve (AUC) used to assess predictive performance. The Youden index (sensitivity + specificity − 1) was applied to identify optimal thresholds. AUC 95% confidence intervals were calculated using the DeLong method, and p-values for AUC were obtained to evaluate statistical significance.

All statistical tests were 2-tailed, and p-values < 0.05 were considered statistically significant. Analyses were performed using IBM SPSS Statistics for Windows, version 25 (IBM Corp., Armonk, N.Y., USA).

Results

A total of 767 patients undergoing coronary angiography were included in the study. Baseline characteristics reflected a representative cardiovascular risk profile. Among the cohort, 496 patients had significant CAS (≥ 50%), while 271 did not (< 50%) (Table 1). Notably, 21.4% had a prior diagnosis of CAD, while 35.6% and 31.7% had hypertension and diabetes mellitus, respectively — comorbidities known to influence inflammatory marker levels. Most patients (96.2%) were in sinus rhythm and had preserved left ventricular ejection fraction (mean 59.4%), indicating overall s cardiac function.

Comparative analysis between the CAS and nonCAS groups revealed significant differences (Table 1). Patients with ≥ 50% CAS had a markedly higher prevalence of established CAD (71.3% vs. 27.8%), reinforcing the progressive nature of

Table 1 – Comparison of patient characteristics according to the presence of significant coronary artery stenosis on coronary angiography

Patient characteristics	Sample (n = 767)	No significant CAS (< 50%) (n = 271)	Significant CAS (≥ 50%) (n = 496)	p-value
Sex (male)	435 (56.7%)	154 (56.8%)	281 (56.7%)	0.26
Age (years)	58.3 ± 12.4	57.7 ± 12.4	58.7 ± 12.4	0.28
BMI (kg/m^2)	26.7 ± 5.6	26.3 ± 4.1	27.1 ± 5.3	0.32
Previous CAD	164 (21.4%)	47 (17.3%)	117 (23.6%)	0.04
Hypertension	273 (35.6%)	91 (33.6%)	182 (36.7%)	0.39
Diabetes mellitus	243 (31.7%)	77 (31.7%)	166 (33.5%)	0.15
Smoking (active)	345 (45.0%)	125 (46.1%)	220 (44.4%)	0.64
Dyslipidemia	139 (18.1%)	49 (18.0%)	90 (18.1%)	0.21
COPD	36 (4.7%)	14 (5.0%)	22 (4.4%)	0.67
Sinus rhythm on ECG	738 (96.2%)	262 (96.4%)	476 (96.0%)	0.11
LVEF (%)	59.4 ± 8.7	60.0 ± 6.9	59.3 ± 7.1	0.41
WBC (10^3/μL)	7.3 ± 2.2	7.2 ± 2.0	7.4 ± 2.4	0.26
Creatinine (mg/dL)	0.8 (0.6-1.2)	0.8 (0.5-1.1)	0.8 (0.6–1.3)	0.68
Hemoglobin (g/dL)	13.5 ± 2.1	13.7 ± 3.1	13.4 ± 1.8	0.25
Neutrophils (10^3/μL)	4.6 (3.7-5.8)	4.57 (3.6-5.6)	4.71 (3.9-6.0)	0.31
Lymphocytes (10^3/μL)	2.1 (1.7-2.7)	2.0 (1.4-2.6)	2.1 (1.8-2.8)	0.52
Platelets (10^3/μL)	252 (213-297)	244 (202-285)	255 (215-301)	0.84
CRP (mg/dL)	0.2 (0.2-0.5)	0.13 (0.09-0.42)	0.44 (0.21-0.82)	0.01
LDL-C (mg/dL)	115.8 ± 35.5	116.4 ± 34.8	115.6 ± 36.0	0.84
HDL-C (mg/dL)	44.0 ± 10.6	43.9 ± 10.9	44.0 ± 10.5	0.84
Triglycerides (mg/dL)	142 (96-204)	144 (98-202)	142 (92-190)	0.65
NLR	2.1 (1.6-3.0)	2.0 (1.5-2.8)	2.6 (1.8-3.6)	0.05
PLR	120.2 (94.1-155.8)	118.6 (91.4-150.8)	129.4 (99.3-161.7)	0.13
AIP	0.52 (0.31-0.70)	0.50 (0.29-0.68)	0.52 (0.32-0.72)	0.78
SII	720.6 (395.1-793.1)	678.8 (372.5-745.7)	742.4 (414.1-815.4)	0.21

AIP: atherogenic index of plasma; BMI: body mass index; CAD: coronary artery disease; CAS: coronary artery stenosis; COPD: chronic obstructive pulmonary disease; CRP: C-reactive protein; ECG: electrocardiography; HDL-C: high density lipoprotein cholesterol; LDL-C: low-density lipoprotein cholesterol; LVEF: left ventricular ejection fraction; NLR: neutrophil-to-lymphocyte ratio; PLR: platelet-to-lymphocyte ratio; SII: systemic immune-inflammation index; WBC: white blood cell count. Data are presented as mean ± standard deviation, median (interquartile range), or n (%), as appropriate.

atherosclerotic disease. Among inflammatory markers, CRP showed the most pronounced elevation in patients with CAS, while NLR was also significantly higher, though to a lesser extent. Other indices — PLR and SII — demonstrated limited discriminatory value, suggesting that CRP and NLR may offer more clinical utility for risk stratification.

Regression analyses provided further insights (Table 2). While univariable analysis identified several potential predictors of CAS, multivariable logistic regression confirmed only prior levels of CAD and CRP as independent predictors.

ROC curve analysis was used to determine diagnostic thresholds (Figure 1). A CRP cutoff > 0.205 mg/dL and NLR > 2.1 showed statistically significant, albeit modest, predictive value. These thresholds may support clinical application while acknowledging their limited standalone utility. AUC values — 0.526 for CRP and 0.510 for NLR — indicate marginal discriminative ability, which reinforces their role as adjunctive tools rather than primary diagnostic tests.

Sex-based subgroup analysis revealed important differences in biomarker profiles (Figure 2). Women had significantly higher PLR values, which may reflect sex-related variations in platelet biology. Conversely, men had higher AIP values, consistent with known sex-specific patterns in lipid metabolism. These findings highlight the relevance of incorporating sex-specific considerations when interpreting cardiovascular biomarkers in clinical practice.

Table 2 – Univariable and multivariable logistic regression analyses for predictors of significant coronary artery stenosis on angiography

Patient characteristics	Univariable logistic regression			Multivariable logistic regression		
	OR	95% CI	p-value	OR	95% CI	p-value
Age (years)	1.007	0.995-1.019	0.28	0.999	0.985-1.014	0.93
Sex (men)	1.007	0.747-1.358	0.96	0.963	0.704-1.315	0.81
Previous CAD	1.471	1.010-2.144	0.028	1.601	1.073-2.128	0.037
Hypertension	1.146	0.840-1.565	0.39	0.939	0.648-1.363	0.74
Diabetes mellitus	1.267	0.917-1.751	0.15	1.197	0.827-1.733	0.34
Smoking (active)	0.931	0.692-1.253	0.64	0.941	0.697-1.271	0.69
CRP	1.188	0.995-1.419	0.044	1.193	1.002-1.385	0.049
NLR	1.081	1.004-1.167	0.048	1.047	0.906-1.210	0.13
PLR	1.023	1.012-1.034	0.037	1.001	0.998-1.004	0.06
AIP	1.099	0.746-1.452	0.87	0.969	0.550-1.710	0.92
SII	0.926	0.539-1.593	0.78	1.008	0.670-1.345	0.70

AIP: atherogenic index of plasma; CAD: coronary artery disease; CRP: C-reactive protein; NLR: neutrophil-to-lymphocyte ratio; OR: odds ratio; PLR: platelet-to-lymphocyte ratio; SII: systemic immune-inflammation index.

Discussion

This study investigated the predictive value of inflammatory and atherogenic biomarkers for identifying CAS in patients undergoing coronary angiography for suspected CAD. It also examined sex-related differences in biomarker expression to assess their potential relevance in risk stratification. The findings indicate that CRP and NLR were modestly associated with the presence of CAS, whereas other markers — including PLR, SII, and AIP — did not exhibit significant predictive value (Central Illustration).

Our results are consistent with some prior studies, while diverging from others, particularly regarding the diagnostic performance of inflammatory biomarkers. Some investigations have reported modest associations between CRP or NLR and angiographically confirmed CAS, with minimal value for PLR and related indices.[11] In contrast, other studies have found stronger associations, which may reflect differences in study design, populations, and endpoint definitions.[12] This heterogeneity highlights the need for cautious interpretation of biomarker data in clinical decision-making.

Among the biomarkers assessed, CRP — a well-established indicator of systemic inflammation — was significantly increased in patients with CAS[13] and remained independently associated with CAS in multivariable analysis. Increased levels of CRP reflect heightened vascular inflammation and have been consistently linked to the pathogenesis of atherosclerosis.[4] However, Mendelian randomization studies, such as those by Ridker et al.,[4] suggest that CRP may act more as a bystander than a causal factor in atherosclerosis. In line with this, the diagnostic performance of CRP in our cohort was poor, with an AUC of 0.526 — only marginally above random chance.[14,15] Despite statistical significance, these findings demonstrate that CRP has limited clinical utility as a standalone diagnostic tool for CAS, challenging the assumption that statistical associations necessarily confer predictive value.[16]

NLR, a composite marker derived from routine blood counts, showed a modest but statistically significant association with CAS.[4] Increased NLR reflects systemic inflammation and immune dysregulation, indicating a shift toward a pro-inflammatory state and diminished adaptive immunity.[17] These findings contribute to the growing body of evidence supporting NLR as a potential marker of cardiovascular risk.[5] However, the low AUC values observed underscore its limited standalone predictive value. Further studies are warranted to evaluate whether combining NLR with other biomarkers can enhance its diagnostic performance.[18]

A key observation of this study was the lack of significant predictive value for other inflammatory and atherogenic markers, including PLR, SII, and AIP. These negative findings are clinically relevant, challenging the assumption that all accessible inflammatory indices are useful for CAD risk stratification. While PLR, for example, has been proposed as a marker of thrombotic risk,[19] our data do not support its utility in predicting CAS. Such discrepancies may arise from variations in patient selection, disease severity, or study methodology, which highlights the importance of rigorous validation before clinical implementation.

Interestingly, PLR, SII, and AIP — though previously suggested as markers of cardiovascular risk — did not show significant associations with CAS in this cohort.[13,20] This may be due to variability in these indices across populations or methodological differences in study design.[21] Moreover, the derived nature of these markers, which depend on multiple interacting laboratory parameters, may introduce noise that weakens their association with specific clinical outcomes.[22]

The study also underscored the importance of considering sex-specific differences in biomarker expression. While levels of CRP and NLR were comparable between sexes, women had higher PLR and men had higher AIP.[13] These differences may

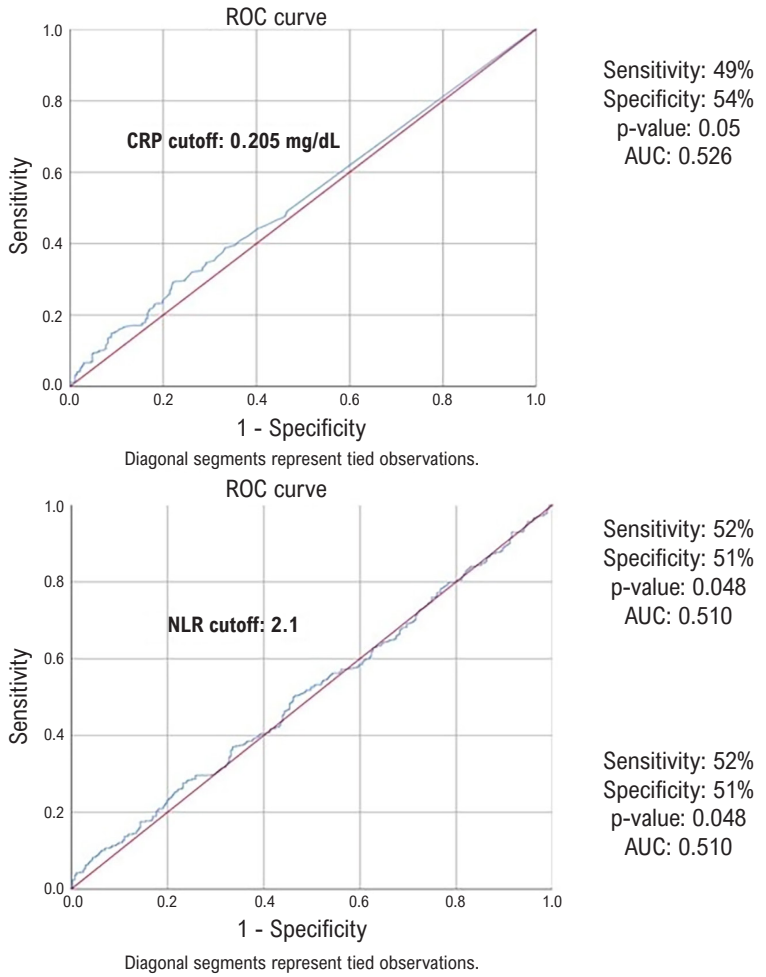

Figure 1 – *CPR: C-Reactive Protein; NLR: Neutrophil-lymphocyte ratio; AUC: Area uder the curve. Receiver operating characteristic (ROC) curve analysis for levels of C-reactive protein (CRP) and neutrophil-lymphocite ratio (NLR) in predicting significant coronary artery stenosis on angiography.*

reflect underlying physiological variations — such as sex-specific immune responses or lipid metabolism — and could influence the interpretation and clinical utility of these markers.[23] For example, increased PLR in females may be linked to heightened thrombogenic potential or immune activity, whereas increased AIP in males could result from unfavorable lipid profiles.[24] Further research is needed to better understand these differences and refine the sex-specific application of biomarkers in assessment of cardiovascular risk.[25]

Despite the contributions of this study to understanding inflammatory and atherogenic markers in CAD, several limitations should be acknowledged. First, its retrospective design introduces potential selection bias and limits causal inference.[26] Second, the analysis did not account for confounders such as medication use, diet, or lifestyle factors that may influence biomarker levels.[27] Third, CAS was assessed by visual estimation during angiography — a method subject to interobserver variability despite its common use.[28] Lastly, the modest AUC values for CRP and NLR reinforce the need for caution in interpreting these markers in isolation. Combining them with advanced imaging or genetic data may improve diagnostic accuracy.[29]

Contrary to previous studies identifying age and male sex as consistent predictors of CAS, our analysis did not find statistically significant associations for these variables. This may be due to the relatively small sample size, which could have limited the statistical power to detect such effects. Additionally, the study population's characteristics — including a narrow age range and potential selection bias toward patients with inflammatory profiles — may have influenced the predictive performance of traditional risk factors.

The study's retrospective design, single-center setting, and limited sample size (767 patients) further constrain generalizability. These factors may have reduced the ability to detect subtle associations, particularly in subgroup analyses or

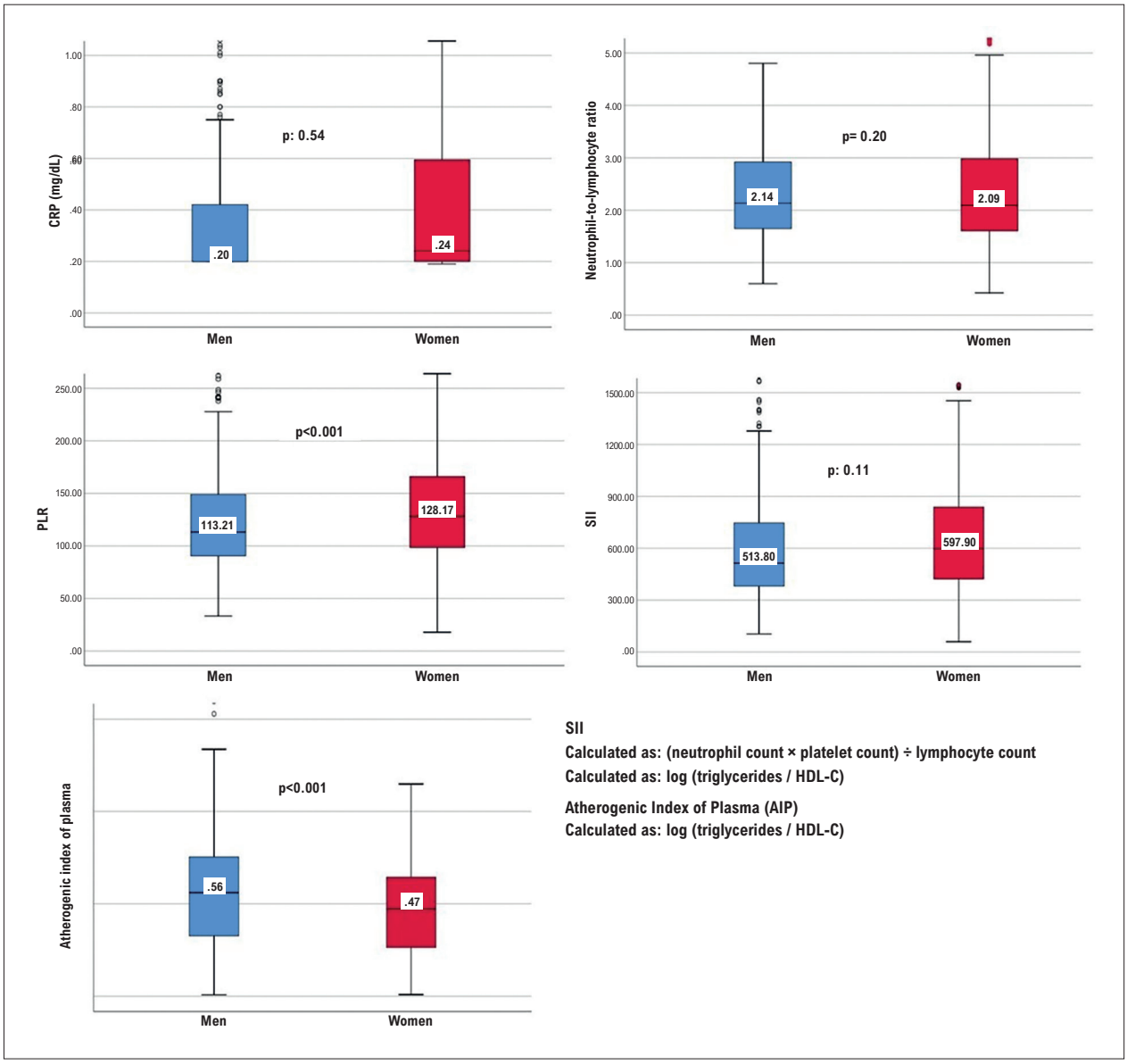

Figure 2 – *Comparison of inflammatory and atherogenic biomarkers between men and women. CRP: C-Reactive protein; NLR: Neutrophil-to-lymphocyte ratio; PLR: Platelet-to-lymphocyte ratio; AIP: Atherogenic index of plasma; SII: Systemic immune-inflammation index. Boxplots illustrate sex-based distributions of C-reactive protein (CRP), neutrophil-to-lymphocyte ratio (NLR), platelet-to-lymphocyte ratio (PLR), atherogenic index of plasma (AIP), and systemic immune-inflammation index (SII) levels. Statistically significant differences were observed for PLR and AIP, with higher PLR in females (p < 0.001) and higher AIP in males (p < 0.001).*

for weaker predictors. Moreover, the study did not adjust for important confounders such as statin use, anti-inflammatory therapies, or comorbidities like chronic kidney disease, all of which could alter biomarker levels. Visual assessment of CAS, though widely used, lacks the precision of techniques such as quantitative coronary angiography or intravascular imaging. These limitations must be considered when interpreting the findings and planning future research.

Importantly, the study's negative findings have meaningful implications for both clinical practice and research. Negative studies serve a critical role in refining our understanding of disease by identifying biomarkers that fail to deliver sufficient predictive performance, despite theoretical promise. Our results clearly show that although inflammatory and atherogenic markers such as CRP and NLR are statistically associated with CAS, their diagnostic performance is poor, with AUC values near random chance. This highlights the need for rigorous clinical validation before integrating biomarkers into routine decision-making. By documenting the limited utility of these widely studied indices, this study helps prevent premature adoption of ineffective diagnostic tools and encourages more efficient use of clinical and research resources.

Conclusion

This study identified significant associations between CAS and both prior CAD and increased levels of CRP. However, the poor predictive performance of CRP and other inflammatory markers — NLR, PLR, SII, and AIP — with diagnostic accuracy only marginally above chance, underscores their limited clinical utility for assessing CAS. These findings suggest that future research should move beyond these commonly studied markers and focus instead on novel biomarkers or integrated, multimodal strategies that combine clinical, biochemical, and imaging data. By clearly demonstrating the limited value of these inflammatory indices, this study contributes to refining the field, helping to redirect efforts and resources toward more promising avenues for effective risk stratification in CAD.

Author Contributions

Conception and design of the research: Can V, Polat F. Acquisition of data: Polat F. Analysis and interpretation of the data: Can V. Statistical analysis: Polat F. Obtaining financing: Can V, Polat F. Writing of the manuscript: Can V. Critical revision of the manuscript for intellectual content: Can V, Polat F.

Potential Conflict of Interest

No potential conflict of interest relevant to this article was reported.

Sources of Funding

There were no external funding sources for this study.

Study Association

This study is not associated with any thesis or dissertation work.

Ethics Approval and Consent to Participate

This study was approved by the Ethics Committee of the Yeni Yüzyıl University under the protocol number 2023/04-1051. All the procedures in this study were in accordance with the 1975 Helsinki Declaration, updated in 2013. Informed consent was obtained from all participants included in the study.

Use of Artificial Intelligence

The authors did not use any artificial intelligence tools in the development of this work.

Availability of Research Data

All datasets supporting the results of this study are available upon request from the corresponding author: Fuat Polat.

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